

University of Michigan

Winter 2024 Instructor Report

EECS 592-001: AI Foundations

Madelyn Gatchel

53 out of 56 students responded to this evaluation.

Responses to University-wide questions about the course:

	SA	A	N	D	SD	N/A	Your Median	School/College Median	Univ-Wide Median
This course advanced my understanding of the subject matter. (Q1631)	18	25	8	1	1	0	4.2	4.4	4.5
My interest in the subject has increased because of this course. (Q1632)	18	17	12	3	3	0	4.0	4.2	4.2
I knew what was expected of me in this course.(Q1633)	23	14	14	1	0	0	4.3	4.4	4.6
I had a strong desire to take this course.(Q4)	17	18	13	5	0	0	4.0	4.0	4.1
As compared with other courses of equal credit, the workload for this course was (SA=Much Lighter, A=Lighter, N=Typical, D=Heavier, SD=Much Heavier). (Q891)	7	12	30	3	0	1	3.3	2.9	3.0

Responses to University-wide questions about the instructor:

	SA	A	N	D	SD	N/A	Your Median	School/College Median	Univ-Wide Median
Madelyn Gatchel seemed well prepared for class meetings.(Q230)	37	11	4	0	1	0	4.8	4.7	4.8
Madelyn Gatchel explained material clearly.(Q199)	35	13	4	0	1	0	4.7	4.6	4.7
Madelyn Gatchel treated students with respect.(Q217)	44	7	2	0	0	0	4.9	4.8	4.8

Responses to questions about the course:

	SA	A	N	D	SD	N/A	Your Median
Overall, this was an excellent course. (Q1)	21	15	9	3	3	0	4.2
I felt included and valued when working with other students. (Q253)	30	16	3	1	1	1	4.7

Responses to questions about the instructor:

	SA	A	N	D	SD	N/A	Your Median
Overall, Madelyn Gatchel was an excellent teacher. (Q2)	33	14	5	0	1	0	4.7

The medians are calculated from Winter 2024 data. University-wide medians are based on all UM classes in which an item was used. The school/college medians in this report are based on classes that are graduate level with enrollment of 16 to 74 in College of Engineering.

Written Comments

Comment on the quality of instruction in this course. (Q900)

Comments
Very good instruction throughout the year!
as I said it was very poorly taught... I know there is a lot to cover in this class but when the teacher cannot apply Bayes rule during the lecture it seems to me like a scandal. Then she said she will come back to us on piazza. Please prepare for the class. A lecturer is not here just to read the notes on the computer. I learned SO MUCH MORE with the book AI, Russel then with the lectures.
All instructors were supportive and very helpful in answering questions – both on Piazza and in office hours. Madelyn was especially helpful – huge thanks to her!
The lectures are at a good pace, very informative, and clear
Overall, pretty well. I thought the material was laid out in a way that was easy to understand. However, there should be more in-class examples, and better worked out examples for understanding. Especially near later half of semester, many of the concepts could have been enriched (sampling, RL, ML)
The quality of instruction is great in this course, where all three instructors (Professor and two GSIs) go over concepts fairly clearly and often engage with students when possible.
Overall the the quality of instruction and communication with students was really good. I constantly was aware of deadlines and the class was organized. However the quality of teaching was not good. I felt the professor and one of the gsi's were just reading off the slides the entire time and when asked with questions, the answers were not satisfying and very often needed getting back to to answer the question. It did not seem like the instructors had a strong grasp of the content and was poorly taught.
Basics can be transferred to one discussion section, and more ML content can be included
Good
I love the course materials! There are technical difficulties sometimes but overall great lectures and discussions
I think that the quality of the instruction, given that it was for an intro class, was ok. I think that we tried to cover too many subjets, and that diluted a lot of the content in the class. For a lot of the hw, we would have to go back and relearn some stuff.
I know this was Professor Bondi-Kelly's first semester teaching, so It wouldn't be appropriate to be harsh on her; but, this was an awful course. The examples, quizzes, research corners, and pretty much everything were super simple, to the point that I felt insulted when we, as PhD students at the University of Michigan, were asked to answer such trivial questions. Overall, the quality of the course just didn't match the quality of the institution that we're all studying at.
I think Prof. Bondi-Kelly is a talented professor who cares about how her students do. She also did a good job of connecting the course material to new and emerging work in the AI field.
I think there can be improvements in the lecture slides, and how new concepts are presented. Often, lecture slides felt more like speaker notes than slides. There was often too much content on each slide, but also a lack of some critical content. For example, acronyms would be used without providing the full phrase, and math notation often was inconsistent between slides.
I found the AIMA textbook very helpful. Often in class, when I could not follow the lecture, I would navigate to the textbook chapter, and was able to follow along.
I feel like I learned a lot during lectures — the instruction quality was really strong.
I loved the enthusiasm of the teaching team!
I think that the lecture material was explained pretty clearly. I also liked solving the problems in the homework and exam, although sometimes, the questions were vaguely worded or seemed detached from the lecture content (ex. HW2 Q2). I would've liked to see more problem examples in class and more content about how the info in class translate to actual code. I felt like I'm wlaing away from this course and still without knowing how to apply many of these techniques in a coding project in a meaningful way.
The teaching staff were excellent, I think I just didn't enjoy the content as much.
Maybe need more detailed instruction on projects but less constraints: I mean there are only constraints on rubric but fewer instruction on how to start a project or say what to choose. It might be better if we could choose from several project topics with detailed expectation or guideline.
The quality of instruction is really good. Prof. Bondi-Kelly gives lots of helps and suggestions to the course project. Both GSI gives clear explanations for questions and materials and great feedbacks for project.
Overall I think the instruction was good, though at times the pacing felt a bit slow. Especially towards the beginning with the earlier topics like search algorithms.
Like this course!
Kind of disorganized. Deadlines kept shifting around, homework descriptions were sometimes too vague and confusing

What were the strengths of the course ? (Q953)

Comments
Teaching and course preparedness.
The textbook
Research corners were helpful in understanding the applications of AI. I also really enjoyed the breadth that the course covered.
Lectures, examples, discussion walkthroughs, take home exam
Research corners
The breadth covered. I really appreciated the foundational exposure across various topics. I feel much more confident in my core understanding of AI
1) This course covers a wide range of topics. 2) The administration of the course is well organized (e.g. logistics, grading, etc) compared with other courses I've taken.
Good organization
Good split of weightage across assignments and project
homework
The research course during the class was helpful.
Discussion were a very big strength of this class as it gave us an opportunity to do more practice problems and really work with the material.
The staffs are all passionate about the subject, and the class itself teaches interesting and helpful topics
Good introduction to foundational topics in AI.
The textbook was amazing. Also, the group project helped me a lot.
The professor put a lot of emphasis on reading new journal articles and conference papers about AI, and discussing them in class. I think this is a good skill to develop for researchers, especially since this field is moving so fast.
Lectures were good, tests were well written and the content will stay with me (but might have been a bit long), and homeworks were similarly fun (the coding sections were especially great).
At the end of the course the transition to walking through how problems were solved was a good change.
Motivating the material with the research corners was interesting.
Hands-on style projects. Lets students take on whatever they want in their own ways.
Instructors were very nice and approachable. They were helpful and receptive to questions during OH and discussions.
The Professor is very enthusiastic.
I think the biggest strength of this course was in the Research Corner section where we would tie the course material directly to current research in the field.
Research corners were fun!

What suggestions would you make for improving the course ? (Q955)

Comments
The pace was fairly slow for the first 2/3 of the class. If we went through some of the basics faster we could have covered more material.
Change the lecturer please.
Clearer discussion explanations
Put less weight and significance on the term project. When this is considered a foundations course, it should be assumed that there is a lower level of expertise in the field of AI. While individual experiences vary, I felt that the term project was a forced component. Given my limited previous work in ML/AI, choosing a project and knowing how to set it up for success was difficult. A course project with expected learnings would be more beneficial and representational of the material than a fully independent research project.
None!
Better teaching/understanding of the course content, less reading off lecture/discussion slides.
Better in-person lectures, more structured content
more examples

Comments
Sometimes, it feels like the professor is just reading out the slide, which makes it hard to follow and understand. AI course involves lots of math, and most math classes actively use whiteboards or iPads to explain the formulas. Also, explaining the intuitions behind each formula would be really helpful. In the end, I think this course should be more like math classes in the university, and should explain further about formulas and theories.
More examples and practice problems in class. I would go so far as to say it would even be worth removing the papers and some of the research corner stuff if necessary to add in practice problems as I felt them to be the most useful in furthering my understanding of the material.
There can be better setup for problem walkthroughs
Project was really frustrating — instructions wanted lots of detail and we got deductions for not elaborating enough, but 5 pages was not nearly enough to explain everything in sufficient detail and include figures/tables. Grading felt overly harsh on many of the project assignments.
I would add more coding/derivations instead of just having theoretical exercises.
Focusing on more advanced topics, with more maths involved would be great. Also, it would be better if Professor Bondi–Kelly could make more eye contact with the students, instead of reading from the slides. For instance, when she started walking through some examples on the whiteboard, that was the best part of the class – so I'd like to see more of that.
I'll restate what I noted above: I think there can be improvements in the lecture slides, and how new concepts are presented. Often, lecture slides felt more like speaker notes than slides. There was often too much content on each slide, but also a lack of some critical content. For example, acronyms would be used without providing the full phrase, and math notation often was inconsistent between slides.
Additionally, the work load for the class felt imbalanced. There was a lot of downtime at the beginning of the semester. Our first HW was due in mid–Feb. In the second half of the semester, we had a midterm, another homework, a project update, project slides, and then a project final report.
I would have liked quicker feedback on projects to give the students more time to implement changes. It would also be useful to have in person meetings about the progress reports instead of just note card comments on the pages.
I found that not enough of the lecture time was spent on the material. Too much was spent on activities and group meetings. The logistical overhead of the non–instructional was not justified by their value. Reading and commenting on papers may be valuable, especially for research oriented students, but I think it is better suited to be a homework assignment.
Working through problems on the Ipad (in landscape) instead of using slides may be a good approach for explaining the algorithms.
The prevalence of rubrics made the class feel very paternalistic. I understand the appeal of them but I worry that the cause students (myself included) to minimize there effort to check boxes instead of working on the totality of the project.
Generally I think the expectations of the students can be raised and more material could be covered in greater depth.
Having lectures and projects being more related.
Out of the research projects that I have done for 500–level classes, the one for this class would probably be my least favorite. I don't understand how we are supposed to come up with a project idea when all we were taught at that point was rational agents and search techniques. This made the learning curve extremely steep, especially when other team members had experience in AI, while I didn't. I felt like I was doing the project blind and struggled to ask for help because I didn't know where to start or even what to ask.
Less weight on project if it's not carefully organized.
I think less project meeting time early on would be helpful. In my case those meetings were so short they didn't really mean anything compared to the weekly meetings we would schedule on our own outside of class.
Work out more examples that align with homework problems in class. Discussions did an okay job of this, more than lecture. Lecture material felt too generic/out of place to be helpful with assignments. Slide decks were disorganized and very hard to use as reference material asynchronously later.

How might the class climate be made more inclusive of diverse students? (Q910)

Comments
N/A very inclusive
Honestly, these kind of questions are ridiculous. More inclusive of diverse students ? We are here to LEARN SOMETHING, the lecturer is not capable of applying Bayes rule in one of the most prestigious universities in this country, I think this should be a matter of worry instead of wondering how to improve the inclusivity which honestly nobody cares about even students from minority groups from which I am part of. So PLEASE stop focusing on things NO STUDENT cares and review the lecture content.
N/a
When pairing up students in groups, take into consideration what classes/previous skills in AI each member has
Assume less working knowledge when it comes with AI/ML (around the project). Or if doing project work, try to pair based on experience such as grouping 1st year master with a 3rd year phd to allow for better knowledge transfer.
Allow students to watch the lecture live via zoom instead of having to watch the lecture recordings if they can not make it to class in-person.
None! The office hours provide a great channel to address any concerns with instructors.
alredaay good
I love thé class climate!
It is.
I think the professor and GSIs did a good job of being inclusive
The class seems pretty inclusive, it was always nice to have access to the lecture and discussion slides in case I forgot or missed something.
Smaller lecture hall.
As someone with less of a background in AI, I think it would have been difficult to conduct meaningful research having only learned the lecture content. Much of the research projects presented in class relied on skills not taught in this class.
N/A
I feel like the class was pretty inclusive so I don't think this needs improvement.
I think it was already